

Saioa Manzano-Morales

COMPARATIVE GENOMICS · BIOINFORMATICS

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Education

Universitat de Barcelona (UB)

PHD IN GENETICS

Specialization: Bioinformatics

Barcelona, Spain

Sept. 2022 - December 2026

Universidad Politécnica de Madrid (UPM)

MS COMPUTATIONAL BIOLOGY

Specializations: Computational-Systems Biology and Genomics, Computational Biology and Data Science

Madrid, Spain

Sept. 2020 - Sept. 2021

Universidad del País Vasco (UPV-EHU)

BS BIOCHEMISTRY AND MOLECULAR BIOLOGY

Leioa, Spain

Sept. 2015 - July 2019

Research Experience

Barcelona Supercomputing Center - Comparative Genomics Group (2022 - Present)

PREDOCTORAL RESEARCH. ADVISOR: DR. TONI GABALDÓN

"Phylogenomics trace microbial interactions and genome dynamics on the path from Asgard to the Last Eukaryotic Common Ancestor"

Barcelona, Spain

CIB Margarita Salas - Biotechnology for Lignocellulosic Biomass (2021 - 2022)

JAE INTRO FELLOWSHIP. ADVISOR: DR. JORGE BARRIUSO

"Degradation of conventional plastics by microbial consortia"

Madrid, Spain

Centre for Plant Biotechnology and Genomics - Evolutionary Dynamics of Genomes, Viruses and Microbial Populations (2021)

MASTER'S THESIS. CO-ADVISORS: DR. JAIME IRANZO, DR. JAIME HUERTA-CEPAS

"Inference and Comparison of Prokaryotic Pangenomes: Orthology vs Sequence Identity. Comparing the PanX, Eggnog-mapper and Roary Pipelines"

Madrid, Spain

University of The Basque Country - Dept of Immunology, Microbiology and Parasitology (2017-2019)

IKASIKER FELLOWSHIP AND BACHELOR'S THESIS. ADVISOR: DR. AITOR REMENTERÍA

"Phenotypic Characterization of strains of *Aspergillus fumigatus*: effect of fmaA mutations on growth and stress response"

Leioa, Spain

Publications

* Sharing first authorship

Bernabeu, M.*, **Manzano-Morales, S.***, Marcet-Houben M.*, & Gabaldón, T. Gene ancestries reveal diverse microbial associations during eukaryogenesis. *Nature* **656**, 399–405 (2026)

Manzano-Morales, S., Gabaldón T. Phylogenomics of Asgard archaea reveals a unique blend of prokaryotic-like horizontal transfer and eukaryotic-like gene duplication. *Nature Communications* **17**, 5103 (2026)

Bernabeu, M.*, **Manzano-Morales, S.***, & Gabaldón, T. Phylogeny-aware simulations suggest a low impact of unsampled lineages in the inference of gene flow during eukaryogenesis. *Genome biology and evolution* **17(11)**, evaf190 (2025)

Langschied, F., Bordin, N., Cosentino, S., Fuentes-Palacios, D., Glover, N., Hiller, M., Hu, Y., Huerta-Cepas, J., Coelho, L. P., Iwasaki, W., Majidian, S., **Manzano-Morales, S.**, Persson, E., Richards, T. A., Gabaldón, T., Sonnhhammer, E., Thomas,

P. D., Dessimoz, C., & Ebersberger, I. Quest for Orthologs in the Era of Biodiversity Genomics. *Genome biology and evolution* **16(10)**, evae224 (2024)

de la Fuente, I*, **Manzano-Morales, S***, Sanz, D., Prieto, A., & Barriuso, J. Quorum sensing in bacteria: in silico protein analysis, ecophysiology, and reconstruction of their evolutionary history. *BMC genomics*, **25(1)**, 441 (2024)

Bernabeu, M*, **Manzano-Morales, S***, & Gabaldón, T. On the impact of incomplete taxon sampling on the relative timing of gene transfer events. *PLoS biology*, **22(3)**, e3002460 (2024)

Manzano-Morales, S*, Liu, Y*, González-Bodí, S., Huerta-Cepas, J., & Iranzo, J. Comparison of gene clustering criteria reveals intrinsic uncertainty in pangenome analyses. *Genome biology*, **24(1)**, 250 (2023)

Presentations

ORAL PRESENTATIONS

Manzano-Morales, S, Gabaldón, T., 21-23 January 2026. *Pervasive horizontal transfer and extensive gene duplication shape Asgard archaeal genomes*: X Meeting of the Spanish Society for Evolutionary Biology, Valencia (Spain). **Oral Presentation**

Manzano-Morales, S, Gabaldón, T., 27 June 2026. *Pervasive horizontal transfer and extensive gene duplication shape Asgard archaeal genomes*: XXIV Jornada de Biología Evolutiva, Barcelona (Spain). **Oral Presentation**

Manzano-Morales, S, Gabaldón, T., 6-8 May 2025. *Pervasive horizontal transfer and extensive gene duplication shape Asgard archaeal genomes*: 12th BSC Doctoral Symposium, Barcelona (Spain). **Oral Presentation**

Manzano-Morales, S, Bernabeu, M., Marcet-Houben M., Gabaldón, T., 14 November 2024. *Diverse ancestries reveal complex symbiotic interactions during eukaryogenesis*: BSC Life Sciences Department Seminar, Barcelona (Spain). **Oral Presentation**

Manzano-Morales, S, Gabaldón, T., 20 June 2024. *Unveiling the relative contribution of horizontal gene transfer and gene duplication in the genome expansion of Asgard archaea*: XXIII Jornada de Biología Evolutiva, Barcelona (Spain). **Oral Presentation**

Manzano-Morales, S, Gabaldón, T., 20 June 2024. *Unveiling the relative contribution of horizontal gene transfer and gene duplication in the genome expansion of Asgard archaea*: XXIII Jornada de Biología Evolutiva, Barcelona (Spain). **Oral Presentation**

Manzano-Morales, S, Bernabeu, M., Marcet-Houben M., Gabaldón, T., 7-8 May 2024. *Reconstructing prokaryotic metabolic contributions to the Last Eukaryotic Common Ancestor*: 11th BSC Doctoral Symposium, Barcelona (Spain). **Oral Presentation**

Manzano-Morales, S, Gabaldón, T., 5 July 2023. *Horizontal Gene Transfer in Asgard Archaea*: XXII Jornada de Biología Evolutiva, Barcelona (Spain). **Oral Presentation**

Manzano-Morales, S, Gabaldón, T., 9-10 May 2023. *Horizontal Gene Transfer in Asgard Archaea*: 10th BSC Doctoral Symposium, Barcelona (Spain). **Oral Presentation**

POSTERS

Manzano-Morales, S, Gabaldón, T., 3 Sept- 5 Oct 2024. *Phylogenomics of Asgard archaea reveals pervasive horizontal gene transfer and a role of gene duplication in expanding genome size*: CGUE, Sant Feliu de Guixols (Spain). **Poster**

Manzano-Morales, S, Gabaldón, T., 17-19 January 2024. *Horizontal Gene Transfer in Asgard Archaea*: SESBE, Málaga (Spain). **Poster**

Manzano-Morales, S, Gabaldón, T., 22-27 July 2023. *Horizontal Gene Transfer in Asgard Archaea*: SMBE, Ferrara (Italy). **Poster**

Manzano-Morales, S, Gabaldón, T., 3-7 October 2022. *A pan-genome view of the Asgard archaea, the closest relatives of eukaryotes*: ERGA Biodiversity Genomics, Online. **Poster & Lightning talk**

Manzano-Morales, S, Gabaldón, T., 17-18 September 2022. *A pan-genome view of the Asgard archaea, the closest relatives of eukaryotes*: Quest for Orthologs 7, Sitges (Spain). **Poster**

Manzano-Morales, S, Gabaldón, T., 12-17 September 2022. *A pan-genome view of the Asgard archaea, the closest relatives of eukaryotes*: EMBO Workshop. Comparative genomics of unicellular eukaryotes: Interactions and symbioses, Sant Feliu de Guixols (Spain). **Poster**

Direction and tutoring

Universidad Internacional de Valencia - Master en Bioinformática

NURIA PLAZA DALMASES

2025

"Orígenes evolutivos de las proteínas implicadas en la meiosis"

Teaching Experience

Universitat Pompeu Fabra (Barcelona) - BS Bioinformatics

COMPARATIVE AND FUNCTIONAL GENOMICS

2021 - 2022 (12h) | 2022 - 2023 (16h) | 2023 - 2024 (16h)

Teaching Assistant

Universitat Pompeu Fabra (Barcelona) - BS Biology

BIOINFORMATICS

2022 - 2023 (10h) | 2023 - 2024 (4h) | 2025 - 2026 (4h)

Teaching Assistant

Awards, Fellowships, & Grants

Grant for Scientific Outreach and Communication, Sociedad Española de Biología Evolutiva, 2025

Prevosti award for the best oral communication, Jornada de Biología Evolutiva - Societat Catalana de Biologia, 2025

IV Bioinformatics Datathon: Best Master's Thesis in Bioinformatics (*not taken*), Universidad de Granada, 2022

Predoctoral Grant (*Finalist*), Gobierno Vasco, 2022

JAE Intro, Consejo Superior de Investigaciones Científicas (CSIC) - 2021-2022

Beca Ikasiker de colaboración, Gobierno Vasco, 2018-2019

Life Sciences Open Day, Barcelona Supercomputing Center, February 2023

BSC Open House, Barcelona Supercomputing Center, September 2022

9th BSC Doctoral Symposium, Barcelona Supercomputing Center, May 2022

Outreach & Professional Development

WORKSHOPS

2024 Workshop in Phylogenomics, Český Krumlov, Czech Republic, 21 January - 3 February 2024

OUTREACH ACTIVITIES

Scientific Outreach in Social Media, Sociedad Española de Biología Evolutiva, 2025 -

ESEB 2025 Social Media, European Society of Evolutionary Biology, 2025

Programa #100tifiques, Fundació Catalana per a la Recerca i la Innovació, February 2025

Crazy about Biomedicine, Fundació Catalunya La Pedrera, January - June 2025

Crazy about HPC, Fundació Catalunya La Pedrera, January - April 2023

Life Sciences Open Day, Barcelona Supercomputing Center, February 2023

BSC Open House, Barcelona Supercomputing Center, September 2022

9th BSC Doctoral Symposium, Barcelona Supercomputing Center, May 2022